**Context Level DFD (0-Level) for eAuto**
**Central Process**
☞ **eAuto Management System**

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◆ **Sub-Processes (high-level inside system)**

1. **Login & Registration** (handles authentication for Customer/Admin)
2. **Product Management** (search, browse, manage product catalog)
3. **Cart & Order Management** (add to cart, checkout, confirm order)
4. **Payment Processing** (verify and confirm transactions)
5. **Supplier & Inventory Management** (stock update, stock requests)
6. **Delivery Management** (assign deliveries, track status)
7. **Reports & Analytics** (sales, stock, delivery reports for Admin)

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◆ **External Entities**

1. **Customer**
2. **Admin**
3. **Supplier**
4. **Delivery Staff**

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◆ **Data Stores**

- **D1: User Database** (stores customer & admin login/registration details)
- **D2: Product Database** (stores product details, categories, pricing, discounts)
- **D3: Cart & Order Database** (stores cart info, order details, order history)
- **D4: Payment/Transaction Database** (stores payment confirmations, transaction logs)
- **D5: Inventory Database** (stores stock levels, supplier updates)
- **D6: Delivery Database** (stores delivery assignments, status updates)
- **D7: Reports Database** (aggregated data for admin reports/analytics)

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◆ **Flow (Data Movement)**

**Customer ↔ eAuto System**

- **Request** → Register/Login, Search Products, Add to Cart, Place Order, Payment, View Orders
- **Response** ← Login Confirmation, Search Results, Order Confirmation, Payment Status, Delivery Status
- **Data Stores Accessed:** User DB, Product DB, Cart & Order DB, Payment DB, Delivery DB

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**Admin ↔ eAuto System**

- **Request** → Manage Products, Manage Orders, Manage Users, View Reports
- **Response** ← Confirmation Messages, Updated Data, Reports
- **Data Stores Accessed:** User DB, Product DB, Order DB, Reports DB, Inventory DB, Delivery DB

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**Supplier ↔ eAuto System**

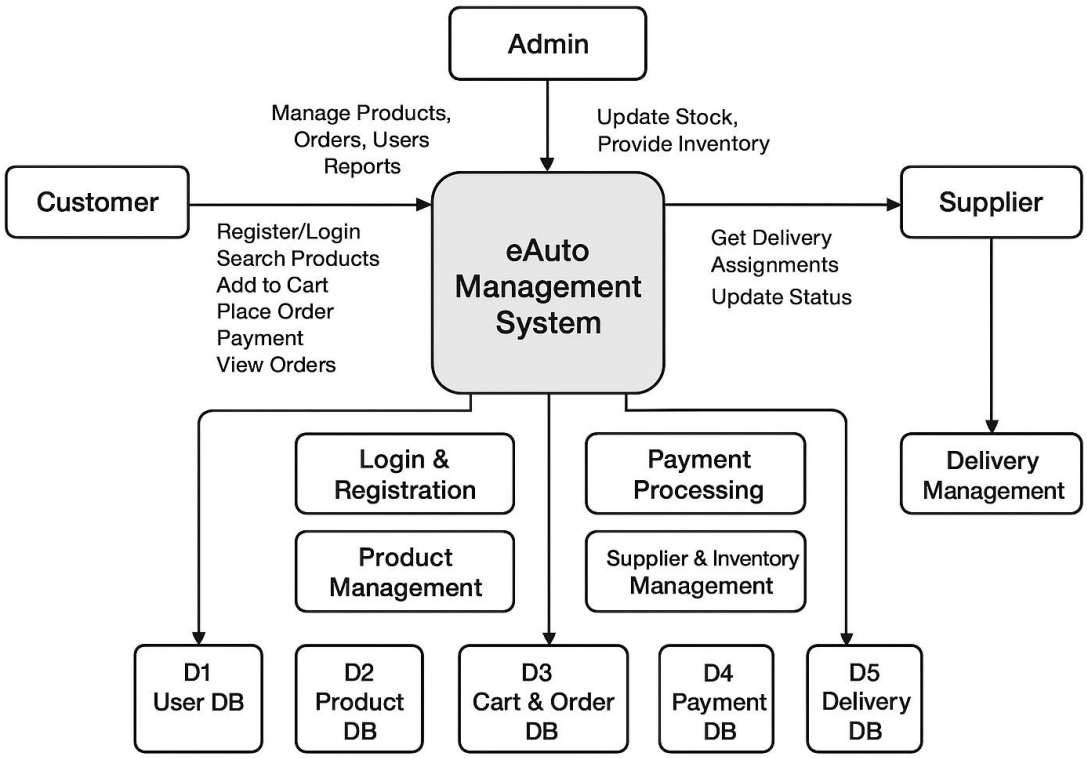
- **Request** → Provide Stock Info, Update Inventory
- **Response** ← Stock Confirmation, Order Requests
- **Data Stores Accessed:** Inventory DB

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**Delivery Staff ↔ eAuto System**

- **Request** → Delivery Assignment, Update Delivery Status
- **Response** ← Delivery Details, Confirmation
- **Data Stores Accessed:** Delivery DB, Order DB

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**1st Level DFD – eAuto Management System**

**Processes (inside the system)**

**1.0 Login & Registration**

- Input: Login/Register request (Customer/Admin)
- Output: Login confirmation
- Data Store: **D1: User Database**

**2.0 Product Management**

- Input: Product search request (Customer), Add/Update product (Admin)
- Output: Product info/search results
- Data Store: **D2: Product Database**

**3.0 Cart & Order Management**

- Input: Add to cart, Place Order (Customer)
- Output: Order confirmation
- Data Store: **D3: Cart & Order Database**

**4.0 Payment Processing**

- Input: Payment details (Customer)
- Output: Transaction confirmation
- Data Store: **D4: Payment Records**

**5.0 Supplier & Inventory Management**

- Input: Stock info update (Supplier)
- Output: Stock confirmation
- Data Store: **D5: Inventory Database**

**6.0 Delivery Management**

- Input: Delivery assignment request (System → Delivery Staff)
- Output: Delivery confirmation, Status update
- Data Store: **D6: Delivery Records**

**7.0 Reports & Analytics**

- Input: Report request (Admin)
- Output: Sales/Stock/Delivery reports

- Data Store: **D7: Reports Database**

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**External Entities**

- **Customer** (Register/Login, Search Products, Add to Cart, Make Payment, Track Delivery)
- **Admin** (Manage Products, Orders, Users, Reports)
- **Supplier** (Stock Updates, Inventory Info)
- **Delivery Staff** (Order Delivery, Update Delivery Status)

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**Data Stores**

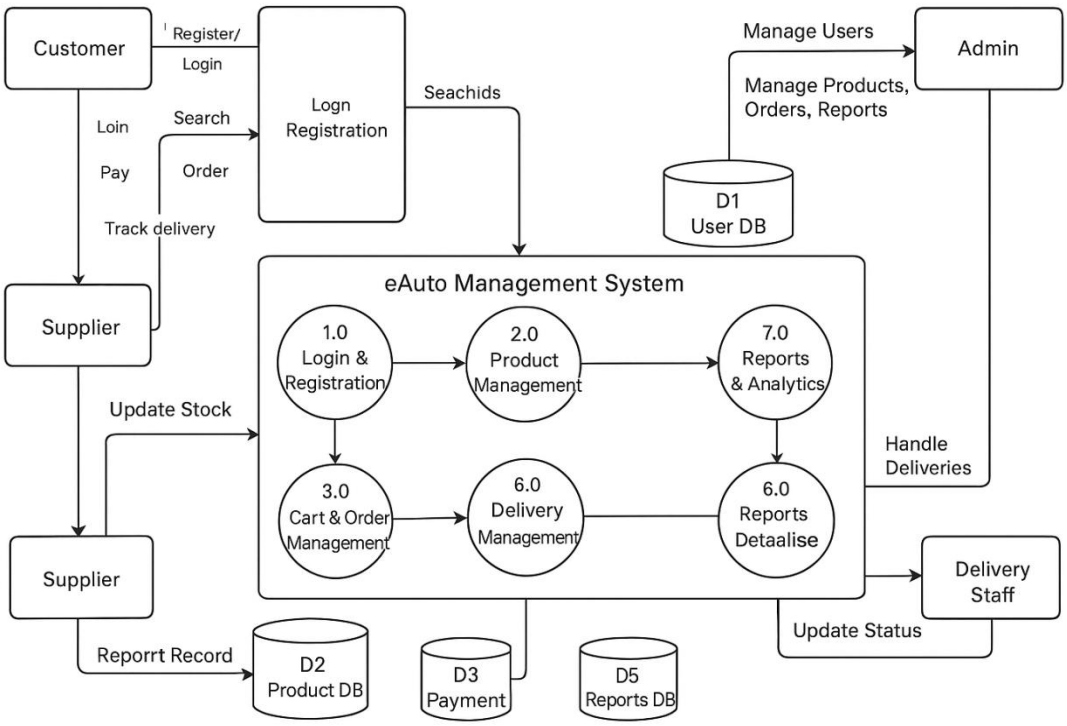
- **D1: User Database** → stores customer/admin login info
- **D2: Product Database** → stores product details
- **D3: Cart & Order Database** → stores cart & order transactions
- **D4: Payment Records** → stores payment details
- **D5: Inventory Database** → manages stock levels
- **D6: Delivery Records** → stores delivery status and assignments
- **D7: Reports Database** → stores analytics data

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**Data Flows**

- Customer ↔ **1.0 Login & Registration** ↔ User Database
- Customer ↔ **2.0 Product Management** ↔ Product Database
- Customer ↔ **3.0 Cart & Order Management** ↔ Cart & Order Database
- Customer ↔ **4.0 Payment Processing** ↔ Payment Records
- Supplier ↔ **5.0 Supplier & Inventory Management** ↔ Inventory Database
- Delivery Staff ↔ **6.0 Delivery Management** ↔ Delivery Records
- Admin ↔ **7.0 Reports & Analytics** ↔ Reports Database

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**1.0 Login & Registration**
**External Entities:**

- **Customer**
- **Admin**

**Data Store:**

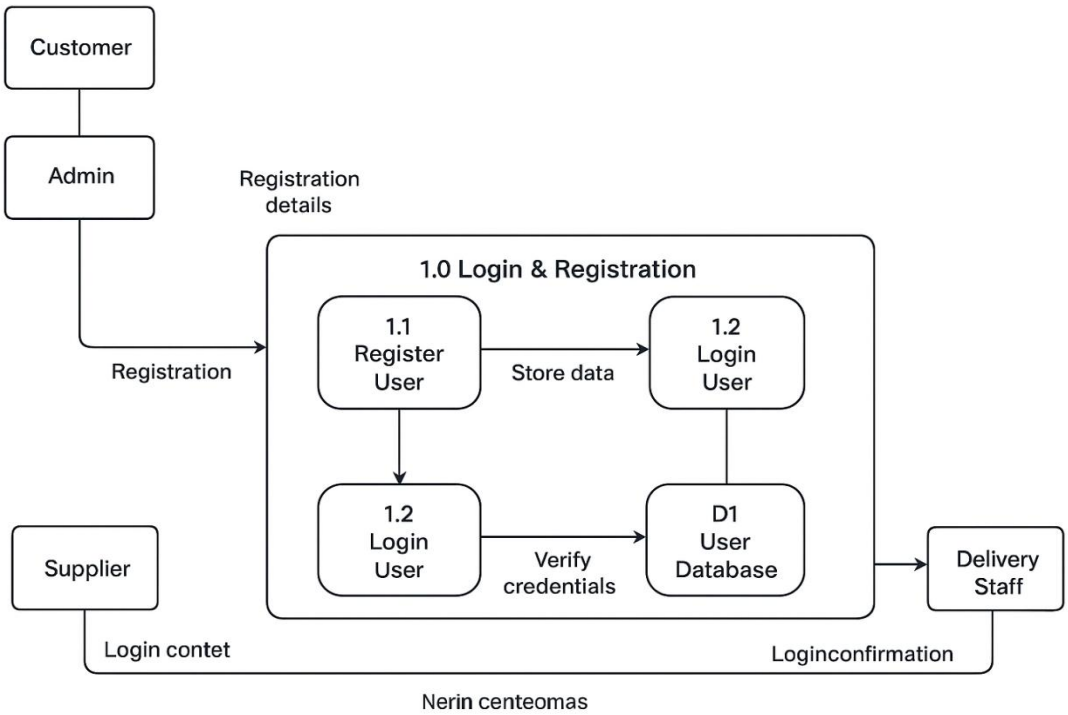
- **D1: User Database**

**Sub-Processes:**

- **1.1 Register User**
  - Input: Registration details (name, email, password, etc.) from Customer/Admin
  - Process: Validate details, store new record
  - Output: Registration confirmation
  - Data Store Interaction: Insert into **D1: User Database**
- **1.2 Login User**
  - Input: Login credentials (username/email & password) from Customer/Admin
  - Process: Validate credentials
  - Output: Login success/failure message
  - Data Store Interaction: Verify credentials from **D1: User Database**

**Data Flows:**

- Customer/Admin → Registration details → 1.1 Register User
- 1.1 Register User → Store data → D1: User Database
- Customer/Admin → Login credentials → 1.2 Login User
- 1.2 Login User → Verify credentials → D1: User Database
- 1.1 Register User → Registration confirmation → Customer/Admin
- 1.2 Login User → Login confirmation/error → Customer/Admin



**2.0 Product Management**

**External Entities:**

- **Customer**
- **Admin**

**Data Store:**

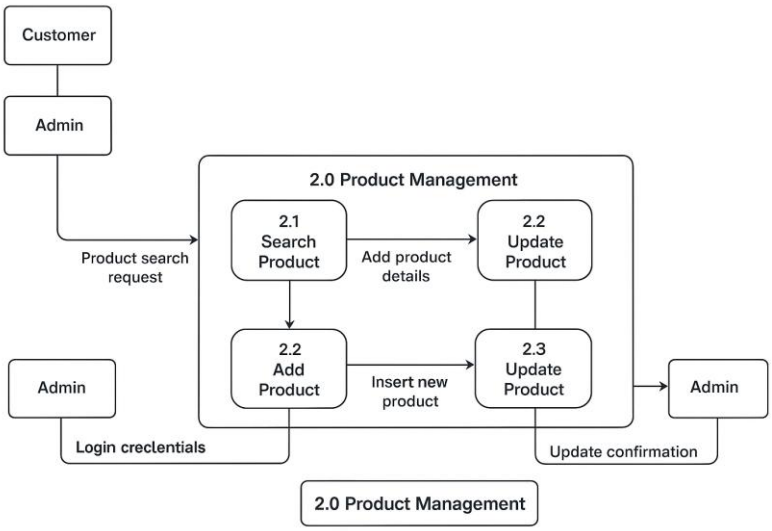
- **D2: Product Database**

**Sub-Processes:**

- **2.1 Search Product**
  - Input: Product search request from Customer
  - Process: Retrieve matching product info
  - Output: Search results (product list/details)
  - Data Store Interaction: Read from **D2: Product Database**
- **2.2 Add Product**
  - Input: New product details from Admin
  - Process: Validate and insert new product
  - Output: Product added confirmation
  - Data Store Interaction: Insert into **D2: Product Database**
- **2.3 Update Product**
  - Input: Updated product details from Admin
  - Process: Validate and update product record
  - Output: Update confirmation
  - Data Store Interaction: Update in **D2: Product Database**

**Data Flows:**

- Customer → Product search request → 2.1 Search Product
- 2.1 Search Product → Fetch product info → D2: Product Database
- D2: Product Database → Product details → 2.1 Search Product
- 2.1 Search Product → Search results → Customer
- Admin → Add product details → 2.2 Add Product
- 2.2 Add Product → Insert new product → D2: Product Database
- 2.2 Add Product → Add confirmation → Admin
- Admin → Update product details → 2.3 Update Product
- 2.3 Update Product → Update record → D2: Product Database
- 2.3 Update Product → Update confirmation → Admin



**3.0 Cart & Order Management**

**External Entities:**

- **Customer**

**Data Store:**

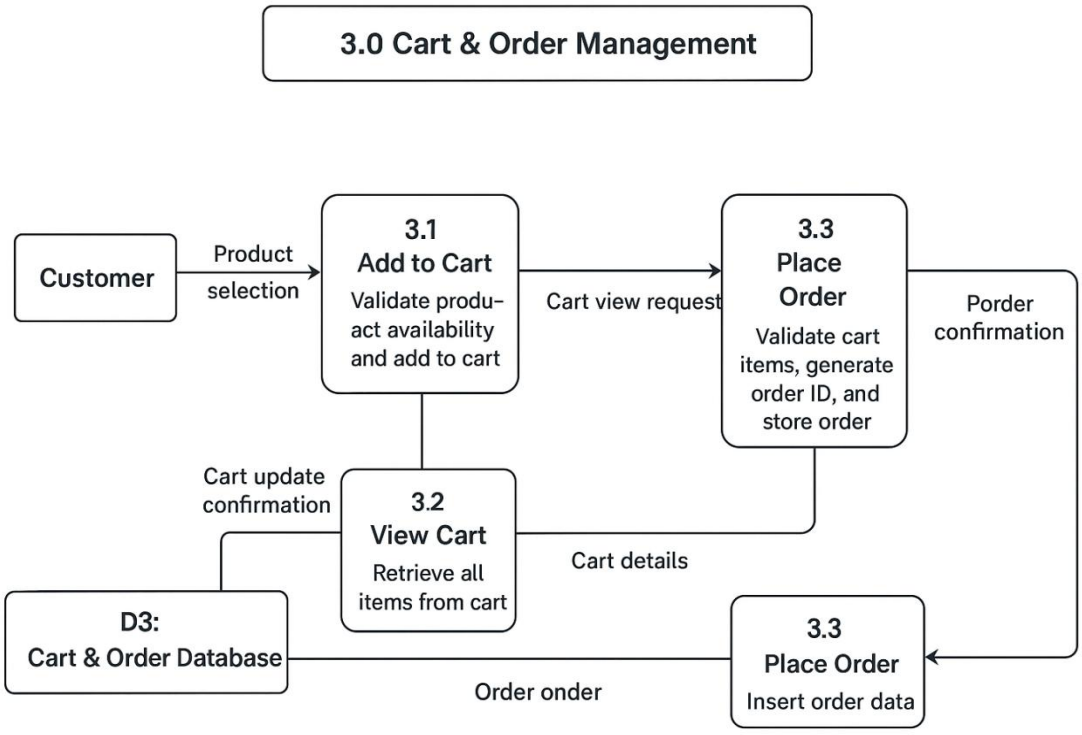
- **D3: Cart & Order Database**

**Sub-Processes:**

- **3.1 Add to Cart**
  - Input: Product selection from Customer
  - Process: Validate product availability and add to cart
  - Output: Cart update confirmation
  - Data Store Interaction: Insert/Update cart record in **D3: Cart & Order Database**
- **3.2 View Cart**
  - Input: Cart view request from Customer
  - Process: Retrieve all items from cart
  - Output: Cart details (products, quantity, total price)
  - Data Store Interaction: Read from **D3: Cart & Order Database**
- **3.3 Place Order**
  - Input: Order request (checkout) from Customer
  - Process: Validate cart items, generate order ID, and store order
  - Output: Order confirmation
  - Data Store Interaction: Insert order details into **D3: Cart & Order Database**

**Data Flows:**

- Customer → Product selection → 3.1 Add to Cart
- 3.1 Add to Cart → Update cart data → D3: Cart & Order Database
- 3.1 Add to Cart → Cart update confirmation → Customer
- Customer → Cart view request → 3.2 View Cart
- 3.2 View Cart → Fetch cart items → D3: Cart & Order Database
- 3.2 View Cart → Cart details → Customer
- Customer → Place order → 3.3 Place Order
- 3.3 Place Order → Insert order data → D3: Cart & Order Database
- 3.3 Place Order → Order confirmation → Customer



**4.0 Payment Processing**

**External Entities:**

- **Customer**
- **Bank/UPI Gateway**

**Data Store:**

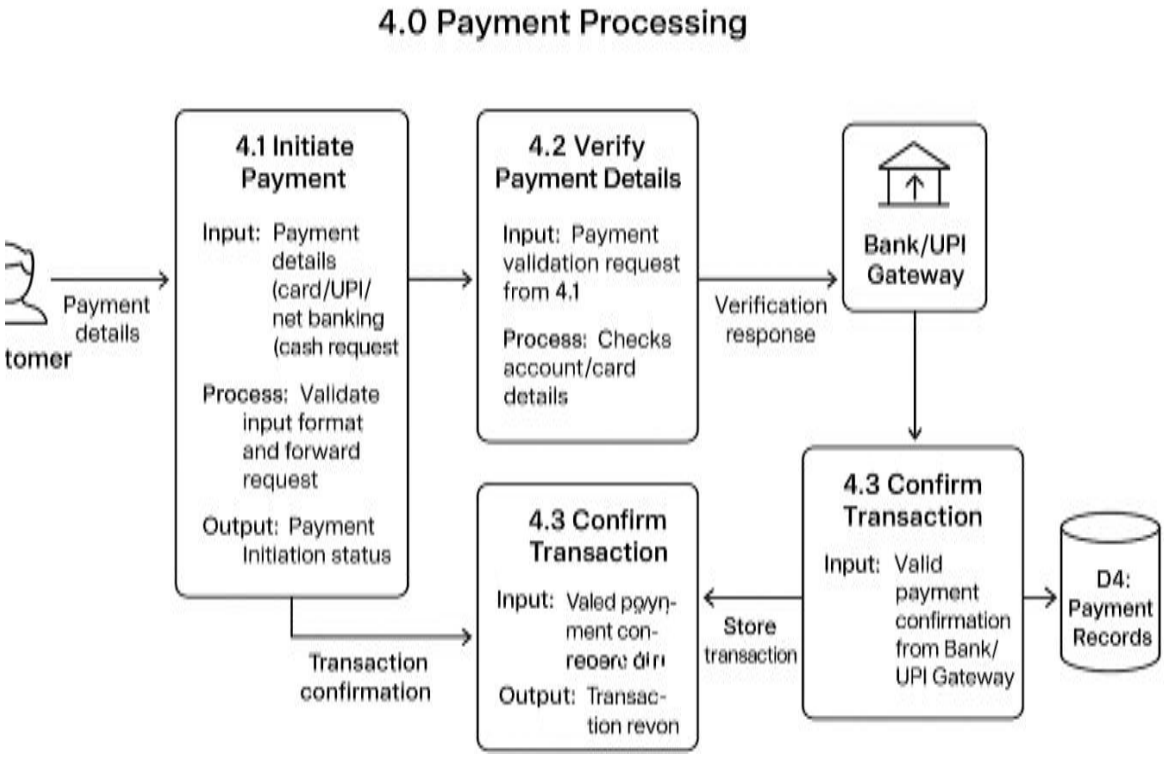
- **D4: Payment Records**

**Sub-Processes:**

- **4.1 Initiate Payment**
  - Input: Payment details (card/UPI/net banking/cash request) from Customer
  - Process: Validate input format and forward request
  - Output: Payment initiation status
  - Data Flow: Send details to **Bank/UPI Gateway**
- **4.2 Verify Payment Details**
  - Input: Payment validation request from 4.1
  - Process: Bank/UPI Gateway checks account/card details
  - Output: Valid/Invalid response
  - Data Flow: Response sent back to **4.2 Verify Payment Details**
- **4.3 Confirm Transaction**
  - Input: Valid payment confirmation from Bank/UPI Gateway
  - Process: Generate transaction record and store in DB
  - Output: Transaction confirmation/Failure message
  - Data Store Interaction: Insert into **D4: Payment Records**

**Data Flows:**

- Customer → Payment details → 4.1 Initiate Payment
- 4.1 Initiate Payment → Forward request → Bank/UPI Gateway
- Bank/UPI Gateway → Verification response → 4.2 Verify Payment Details
- 4.2 Verify Payment Details → Validated result → 4.3 Confirm Transaction
- 4.3 Confirm Transaction → Store transaction → D4: Payment Records
- 4.3 Confirm Transaction → Transaction confirmation → Customer



**5.0 Supplier & Inventory Management**
**External Entities:**

- **Supplier**
- **Admin**

**Data Store:**

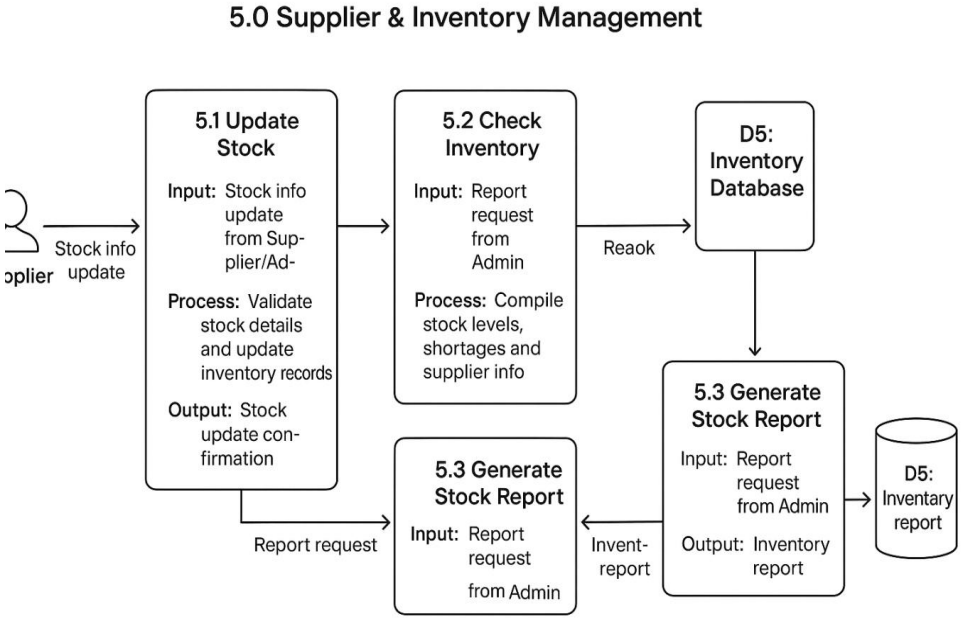
- **D5: Inventory Database**

**Sub-Processes:**

- **5.1 Update Stock**
  - Input: Stock info update from Supplier/Admin
  - Process: Validate stock details and update inventory records
  - Output: Stock update confirmation
  - Data Store Interaction: Update **D5: Inventory Database**
- **5.2 Check Inventory**
  - Input: Inventory check request from Admin/System
  - Process: Retrieve current stock details
  - Output: Stock availability report
  - Data Store Interaction: Read from **D5: Inventory Database**
- **5.3 Generate Stock Report**
  - Input: Report request from Admin
  - Process: Compile stock levels, shortages, and supplier info
  - Output: Inventory report
  - Data Store Interaction: Access data from **D5: Inventory Database**

**Data Flows:**

- Supplier/Admin → Stock info update → 5.1 Update Stock
- 5.1 Update Stock → Update records → D5: Inventory Database
- 5.1 Update Stock → Stock confirmation → Supplier/Admin
- Admin/System → Inventory check request → 5.2 Check Inventory
- 5.2 Check Inventory → Retrieve stock data → D5: Inventory Database
- 5.2 Check Inventory → Stock availability report → Admin/System
- Admin → Report request → 5.3 Generate Stock Report
- 5.3 Generate Stock Report → Access data → D5: Inventory Database
- 5.3 Generate Stock Report → Inventory report → Admin



**6.0 Delivery Management**

**External Entities:**

- **System**
- **Delivery Staff**
- **Customer**

**Data Store:**

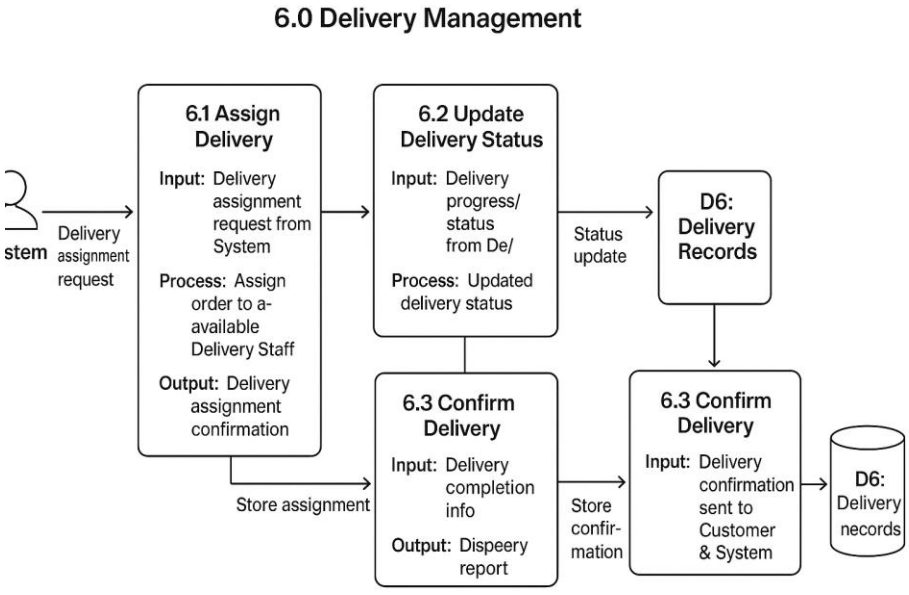
- **D6: Delivery Records**

**Sub-Processes:**

- **6.1 Assign Delivery**
  - Input: Delivery assignment request from System
  - Process: Assign order to available Delivery Staff
  - Output: Delivery assignment confirmation
  - Data Store Interaction: Insert assignment into **D6: Delivery Records**
- **6.2 Update Delivery Status**
  - Input: Delivery progress/status from Delivery Staff
  - Process: Record status updates (Out for delivery, Delivered, Failed)
  - Output: Updated delivery status
  - Data Store Interaction: Update **D6: Delivery Records**
- **6.3 Confirm Delivery**
  - Input: Delivery completion info from Delivery Staff
  - Process: Validate and confirm successful delivery
  - Output: Delivery confirmation sent to Customer & System
  - Data Store Interaction: Store confirmation in **D6: Delivery Records**

**Data Flows:**

- System → Delivery assignment request → 6.1 Assign Delivery
- 6.1 Assign Delivery → Store assignment → D6: Delivery Records
- 6.1 Assign Delivery → Assignment confirmation → Delivery Staff
- Delivery Staff → Status update → 6.2 Update Delivery Status
- 6.2 Update Delivery Status → Update record → D6: Delivery Records
- 6.2 Update Delivery Status → Updated status → System/Customer
- Delivery Staff → Delivery completion info → 6.3 Confirm Delivery
- 6.3 Confirm Delivery → Store confirmation → D6: Delivery Records
- 6.3 Confirm Delivery → Delivery confirmation → Customer
- 6.3 Confirm Delivery → Delivery confirmation → System



**7.0 Reports & Analytics**

**External Entities:**

- **Admin**
- **System**

**Data Store:**

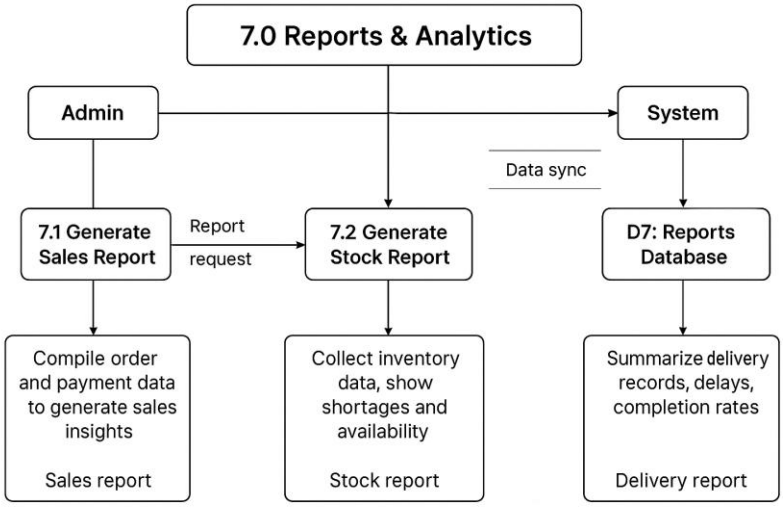
- **D7: Reports Database**

**Sub-Processes:**

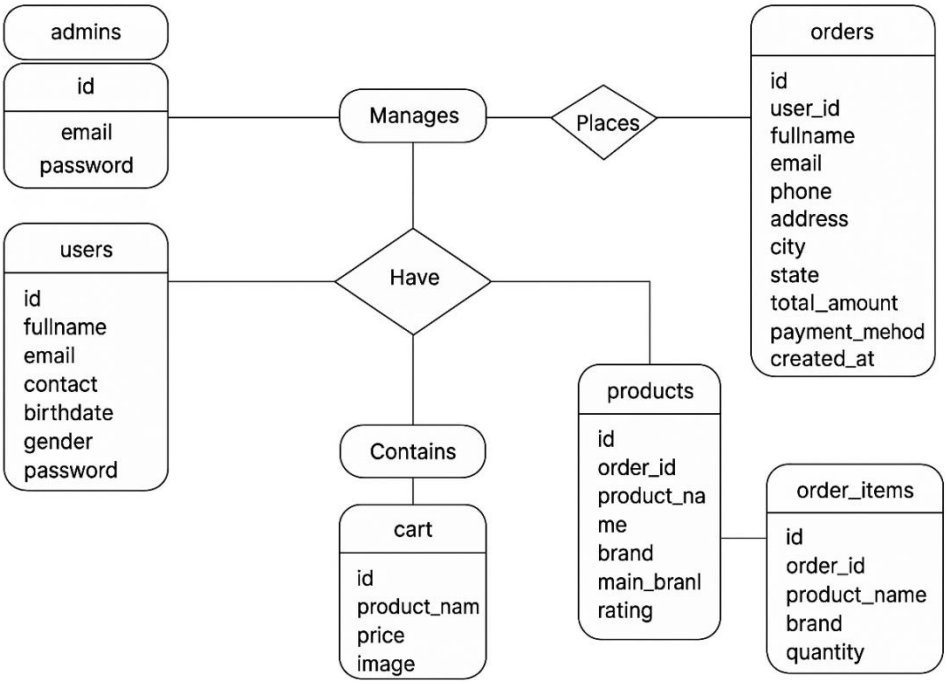
- **7.1 Generate Sales Report**
  - Input: Report request (sales) from Admin
  - Process: Compile order and payment data to generate sales insights
  - Output: Sales report
  - Data Store Interaction: Retrieve/store in **D7: Reports Database**
- **7.2 Generate Stock Report**
  - Input: Report request (stock) from Admin
  - Process: Collect inventory data, show shortages and availability
  - Output: Stock report
  - Data Store Interaction: Retrieve/store in **D7: Reports Database**
- **7.3 Generate Delivery Report**
  - Input: Report request (delivery) from Admin
  - Process: Summarize delivery records, delays, completion rates
  - Output: Delivery report
  - Data Store Interaction: Retrieve/store in **D7: Reports Database**

**Data Flows:**

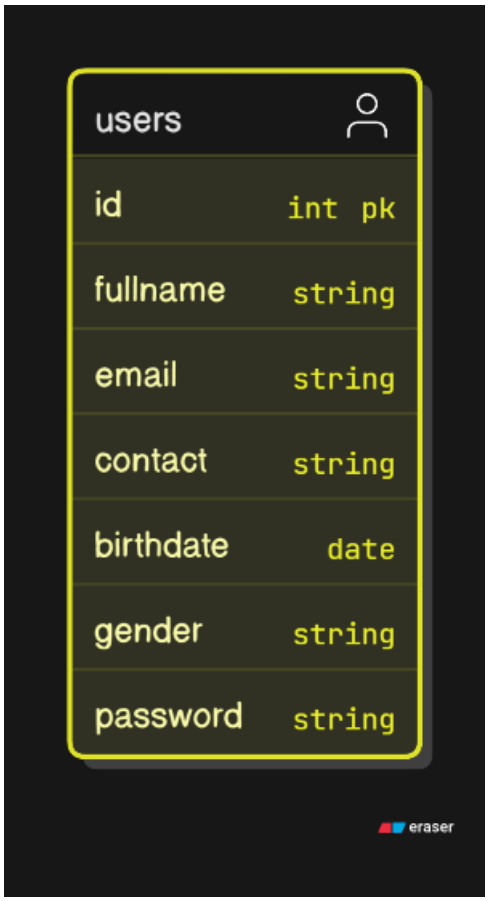
- Admin → Report request → 7.1 / 7.2 / 7.3
- 7.1 Generate Sales Report → Fetch sales data → D7: Reports Database
- 7.1 Generate Sales Report → Sales report → Admin
- 7.2 Generate Stock Report → Fetch inventory data → D7: Reports Database
- 7.2 Generate Stock Report → Stock report → Admin
- 7.3 Generate Delivery Report → Fetch delivery data → D7: Reports Database
- 7.3 Generate Delivery Report → Delivery report → Admin
- System → Data sync (orders, inventory, delivery) → D7: Reports Database

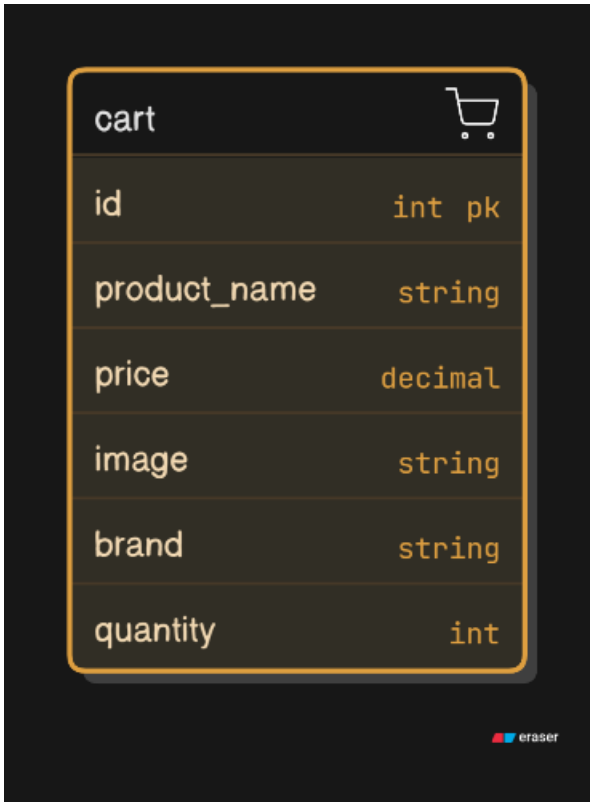


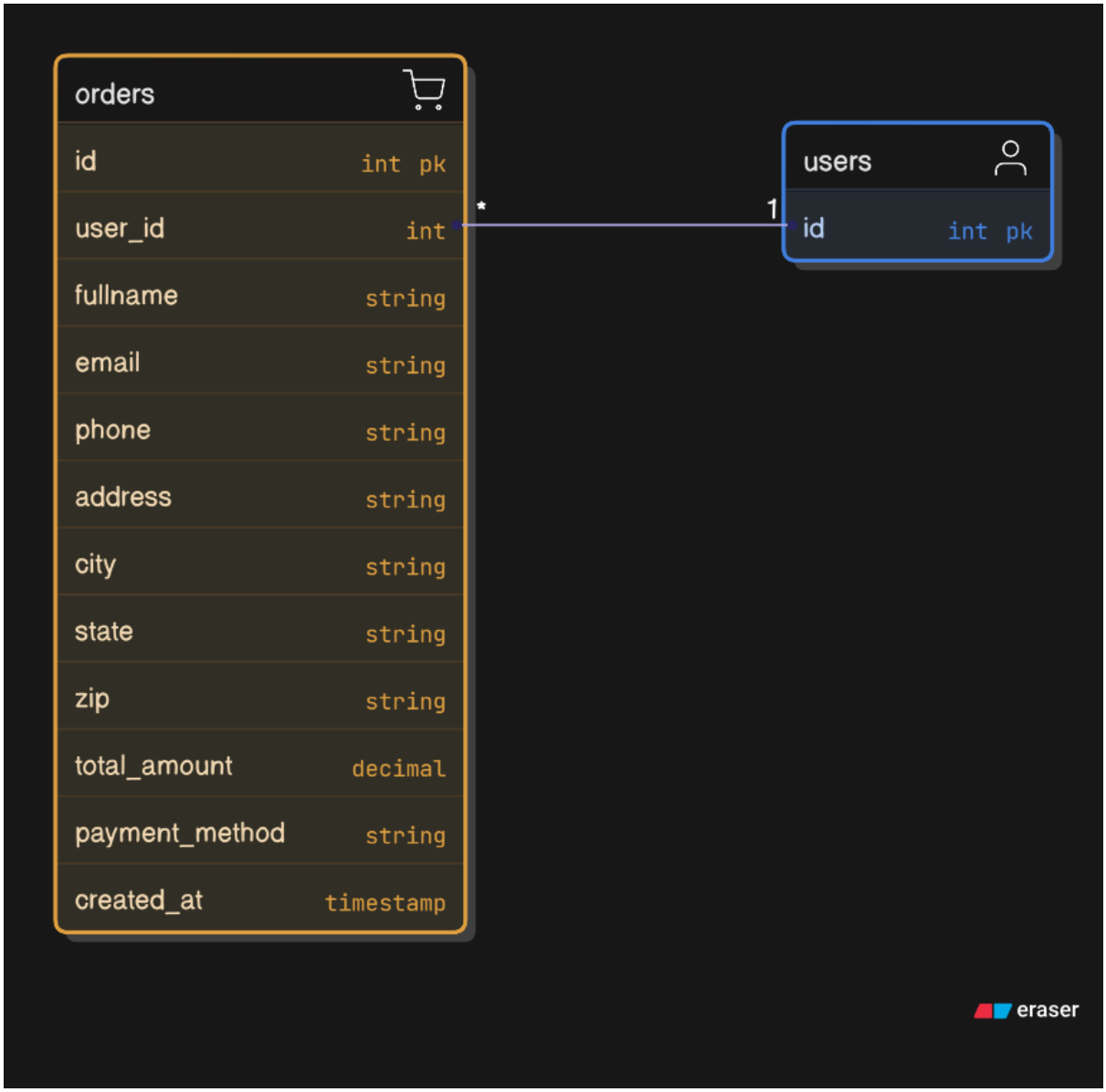
## ER Diagram for eAuto Database

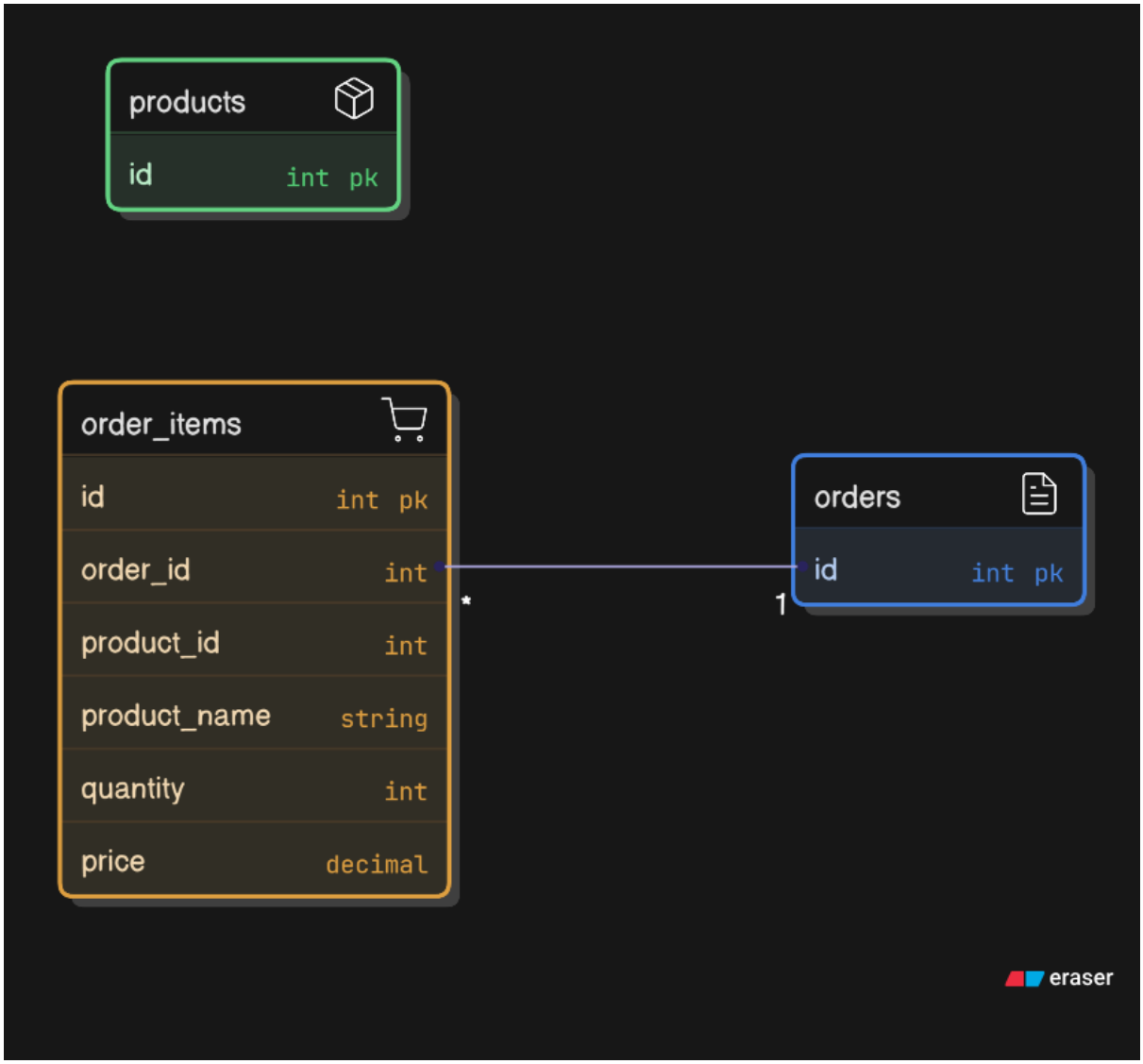


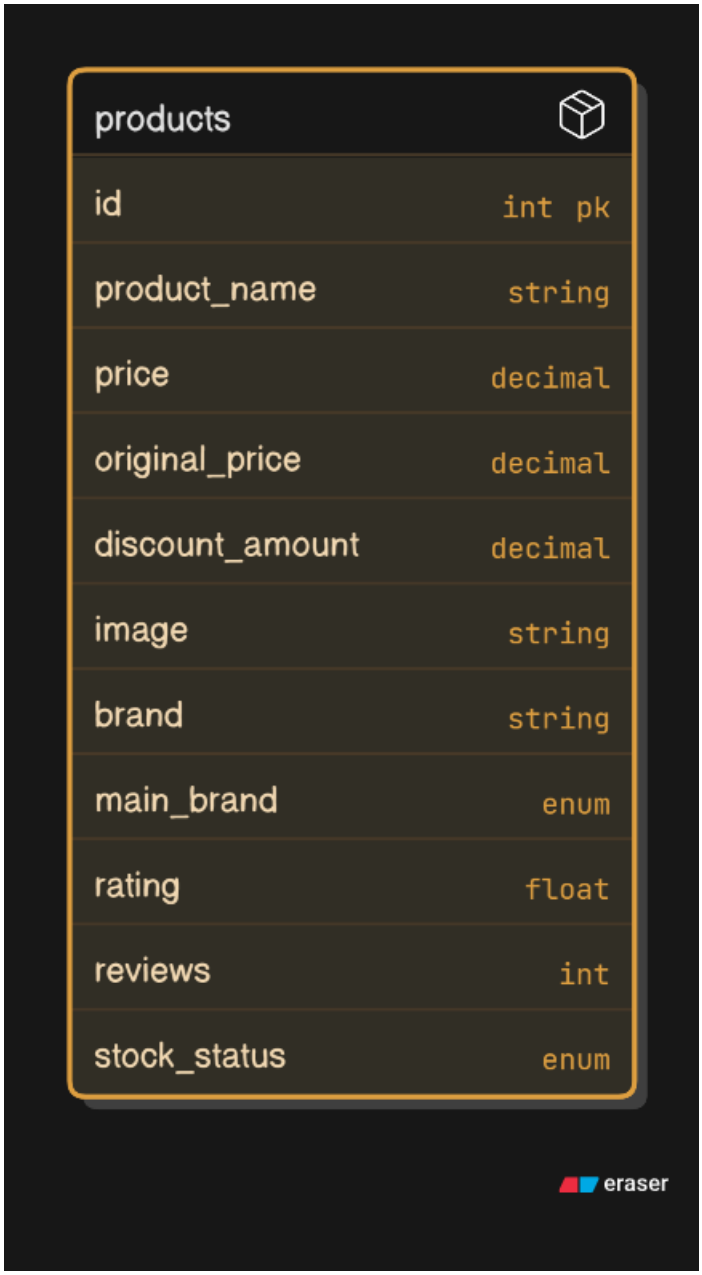

admins
id int pk
email string unique
password string
created_at timestamp
eraser


users
id int pk
fullname string
email string
contact string
birthdate date
gender string
password string
eraser


cart
id
int pk
product_name
string
price
decimal
image
string
brand
string
quantity
int
eraser


orders
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user_id int
fullname string
email string
phone string
address string
city string
state string
zip string
total_amount decimal
payment_method string
created_at timestamp
*
1
users
id int pk
eraser


products
id
int pk
order_items
id
int pk
order_id
int
product_id
int
product_name
string
quantity
int
price
decimal
orders
id
int pk
*
1
eraser


products
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price
decimal
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stock_status
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